Speech Recognition for Sign Language

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Bridging the Communication Gap

Sign language is vital for individuals with hearing or speech impairments, but a lack of general knowledge creates barriers. This project aims to bridge this gap by translating hand gestures into text and speech in real time.

Real-time Translation

Hand gestures to readable text and audible speech.

Leverages CV & ML

Uses OpenCV, MediaPipe, and Scikit-learn for detection and classification.

Accurate & Robust

Voting Ensemble model ensures high accuracy and fast inference.

Addressing the Problem

Individuals relying on sign language often face communication difficulties. Existing solutions are costly or computationally heavy. We need a cost-effective, real-time, camera-based system.

Project Objectives

- Real-time hand gesture recognition via standard webcam.
- Accurate hand landmark tracking using MediaPipe.
- Extraction of invariant geometric features.
- Training and evaluation of ML models for classification.
- Real-time text output and optional speech feedback.

System Overview

Our proposed system uses a vision-based approach with classical machine learning, extracting geometric features from hand landmarks for faster, lower-cost execution.

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Capture Video

Live video input from a webcam.

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Detect Landmarks

MediaPipe identifies 21 key hand points.

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Extract Features

Normalized geometric features are extracted.

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Classify Gestures

Voting Ensemble model predicts the gesture.

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Output & Speech

Display predicted label and generate speech.

Implementation Details

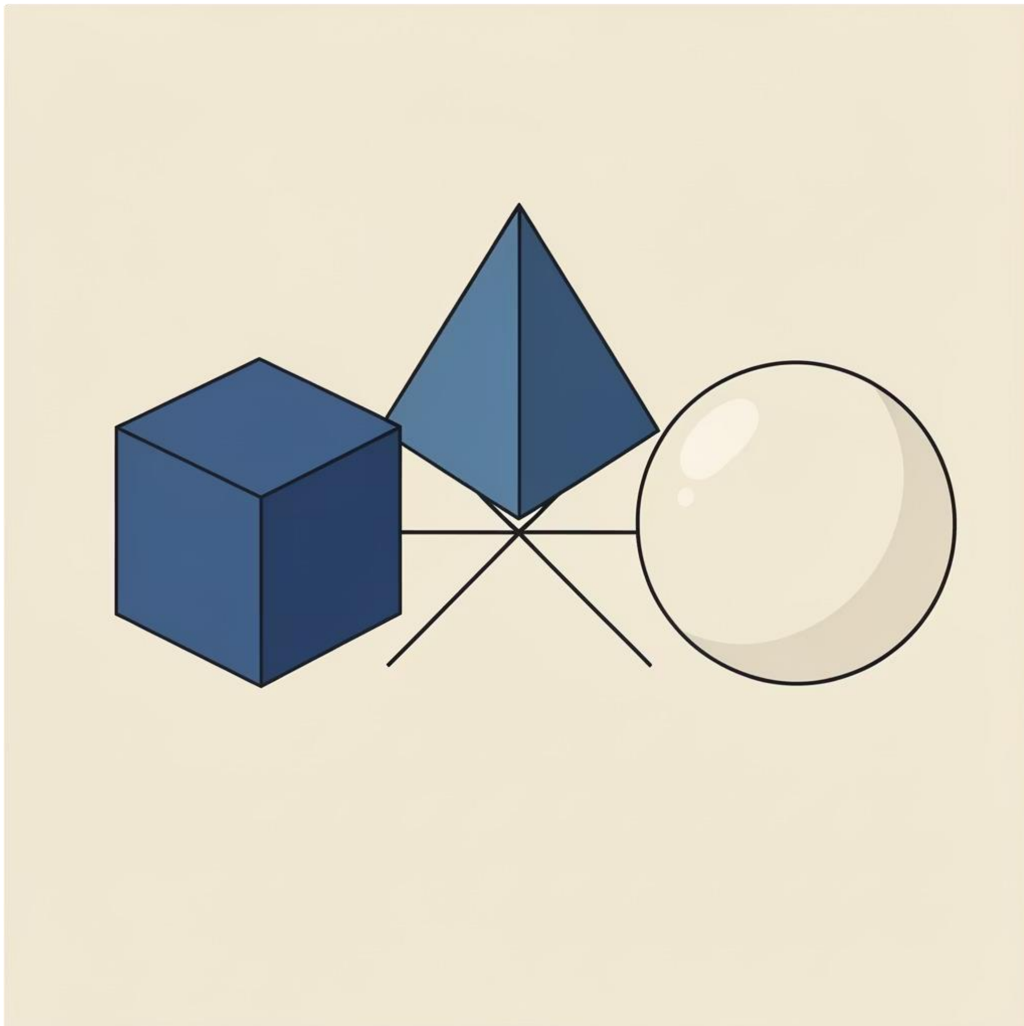
The system uses Python 3.x, OpenCV, MediaPipe, and Scikit-learn. Feature extraction ensures robustness against hand variations.

Feature Extraction

- Normalization relative to wrist joint.
- Scale invariance using max wrist distance.
- Rotation alignment for consistency.
- Feature vector: coordinates, distances, angles.

Classification Algorithm

A Voting Ensemble Classifier combines SVM, Random Forest, and MLP for improved accuracy and stability.



Performance & Requirements

The Voting Ensemble model achieved 96% accuracy, outperforming individual models. The system is designed for standard personal computers.

Model Accuracy

SVM	92%
Random Forest	94%
MLP	93%
Voting Ensemble	96%

System Requirements

- **Hardware:** Intel Core i5, 4GB RAM, Webcam, 500MB storage.
- **Software:** Windows 10/11, Python 3.8+, required libraries.

Conclusion & Future Vision

This project delivers an accessible, efficient, and scalable real-time sign language recognition system.

Successful Implementation

Real-time sign language recognition using CV and ML.

Future Enhancements

Dynamic gesture recognition, mobile app development, expanded vocabulary, and multilingual support.